

PRISM

GRANULAR AMBIENCE & SHIMMER

USER MANUAL

Version 1.0

FREQ DSP

DESIGNED BY BIBAS

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INTRODUCTION

Welcome to PRISM

PRISM turns any sound into a wide, airy, ethereal cloud. Put it on keys, pads, guitars, vocals or synths. It scatters your sound into thousands of tiny pieces, lets part of them climb into shimmering octaves and fifths, and wraps everything in a large, soft space that moves and breathes.



GRAINS

Your sound is cut into tiny overlapping slices called grains and scattered across time and the stereo field.



SHIMMER

Pitched copies climb through the reverb, octave by octave, for a glowing, angelic top layer.



SPACE

A dense 16-line reverb with early reflections, a long soft tail and a ducker that keeps your source clear.

Quick start in five steps

1. **Insert PRISM** on a track as an effect. In Logic Pro: Audio FX slot, Audio Units, FREQ DSP, PRISM. In FL Studio: a mixer insert slot.
2. **Pick a preset** with the arrows or the name menu in the top bar. Start with **Angel Choir** or **Soft Shimmer**.
3. **Play something sustained,** such as a chord or a pad, and set **MIX** to taste.
4. **Shape the glow.** More **SHIMMER** for sparkle, bigger **SIZE** and longer **DECAY** for space, more **BLOOM** for swell.
5. **Polish it.** Use **DARKEN**, **LOW CUT** and **HIGH CUT** in the bottom strip, and **DUCKER** to keep your source clear.

At a glance

FORMATS
Audio Unit (Logic Pro), VST3 (FL Studio), Standalone

SYSTEM
macOS 12 or later, Apple Silicon and Intel

CONTROLS
15 automatable parameters and 3 switches

CHANNELS
Stereo in, stereo out

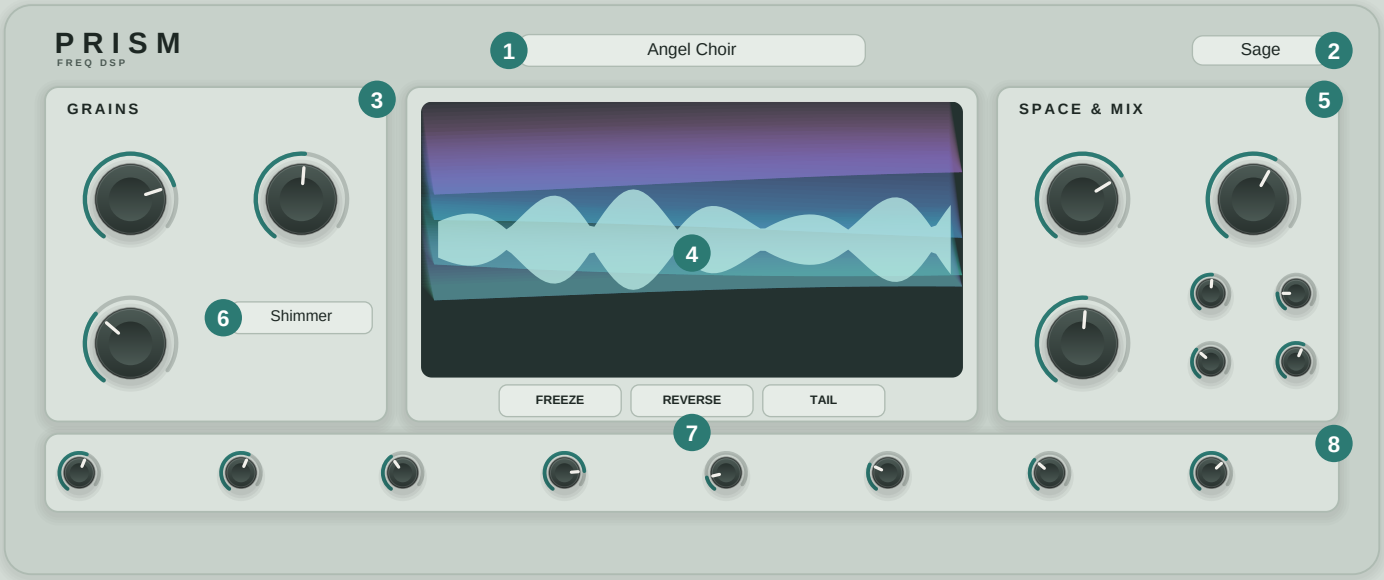
PRESETS
10 factory presets

THEMES
Sage, Pearl and Graphite

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GETTING AROUND

A tour of the interface



1. **Preset browser.** Arrows or the name menu step through the 10 factory presets.
2. **Theme menu.** Sage, Pearl or Graphite. Saved with your project.
3. **GRAINS panel.** Spread, Shimmer and Bloom, plus the Pitch Mode menu (6).
4. **Aurora display.** A glowing waveform ribbon over soft aurora light, with a bloom for every grain.
5. **SPACE & MIX panel.** Size, Decay and Mix, with Halo, Damping, Ducker and Pre-Delay below.
6. **Pitch Mode menu.** Chooses which pitches the shimmer can reach.
7. **Switches.** Freeze, Reverse and Tail sit under the display.
8. **Detail strip.** Grain Size, Density, Low Cut, High Cut, Darken, Mod Rate, Mod Depth and Output.

Handy tricks

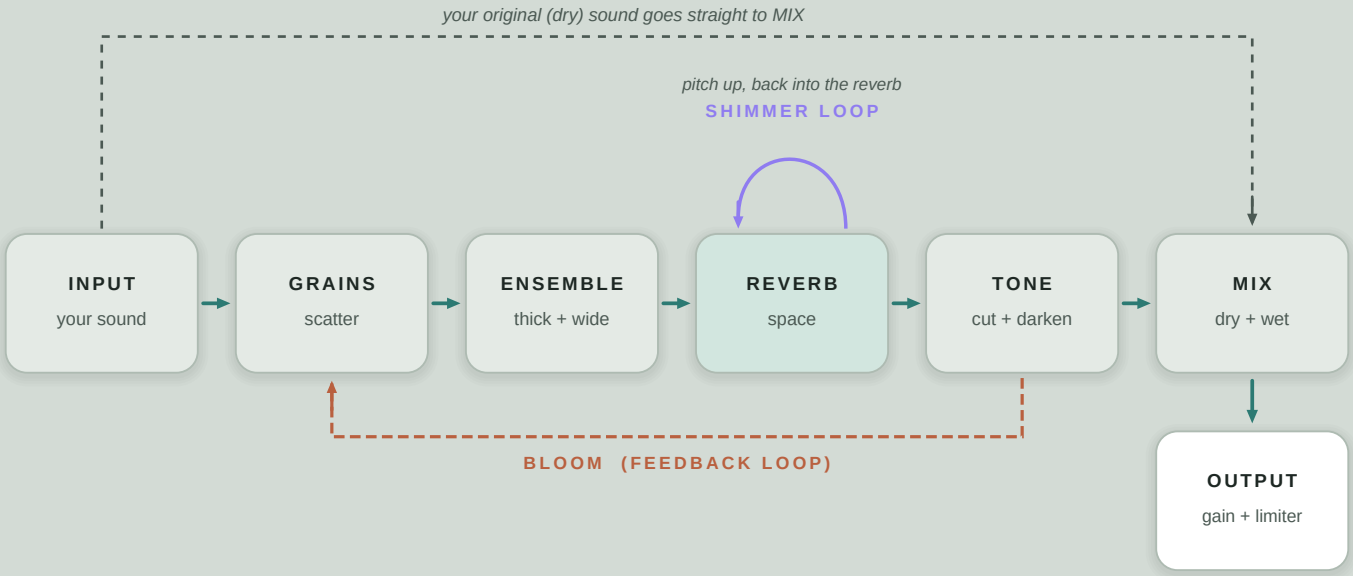
- **Hover** over any control for a short explanation.
- **Drag** a knob up or down, or use the scroll wheel. **Double-click the number** under a knob to type an exact value.
- **Resize** the window by dragging the bottom-right corner. The layout keeps its proportions.
- Every control can be **automated** from your DAW, and everything is saved with the project.

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UNDER THE HOOD

How PRISM works

Everything you send into PRISM follows one path. Knowing it makes every control easier to understand.



What each stage does

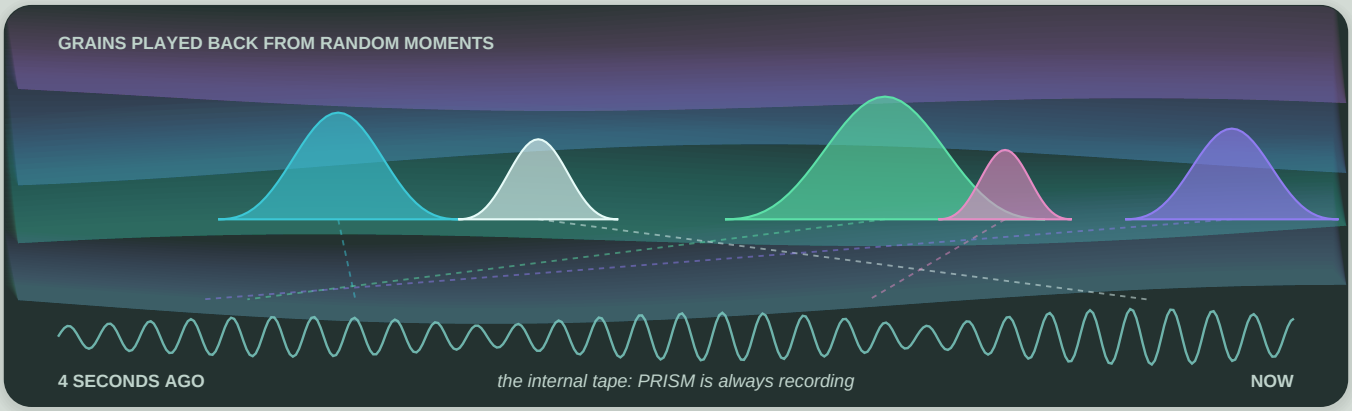
1 GRAINS	PRISM continuously records about four seconds of your sound onto an internal tape, then plays back many short overlapping slices (grains) from random moments on that tape. Every grain fades in and out smoothly, gets a tiny random detune and its own place in the stereo field.
2 ENSEMBLE	Three slowly moving copies per side thicken and widen the cloud, like a high-end chorus.
3 REVERB	A dense 16-line reverb with a pre-delay and early reflections. Part of its output is pitch-shifted and fed back into it: the shimmer loop. Halo, Damping and Ducker shape and tidy the tail.
4 TONE	Low Cut and High Cut remove unwanted lows and highs from the effect. Darken tilts the balance toward warmth. A little extra air and stereo motion are added here.
5 BLOOM	A portion of the finished wet sound is fed back into the grain tape so it swells and smears. It is softly limited and always stays below 100 percent, so it can never run away.
6 MIX AND OUTPUT	MIX blends your sound with the effect at equal power, so the level stays steady. OUTPUT sets the final level, and a gentle safety limiter sits just below full scale.

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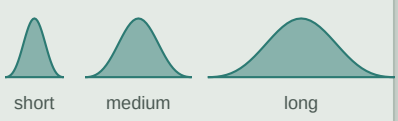
TEXTURE

The grain engine

A grain is a tiny slice of sound, between 10 and 300 milliseconds long. PRISM plays many of them in overlapping layers, evenly spaced so the level stays steady and silky. Three controls decide how the cloud behaves: **Grain Size**, **Density** and **Spread**.

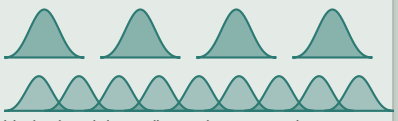


GRAIN SIZE



10 to 300 ms. Short grains are glittery. Long grains are smooth and washy.

DENSITY



thin (top) and dense (bottom): more overlap

0 to 100%. How thickly grains overlap, from 3 to 16 at once. Higher is smoother.

SPREAD



0%: close 100%: far and wide

0 to 100%. How far back in time grains reach, and how wide they scatter.

The aurora display

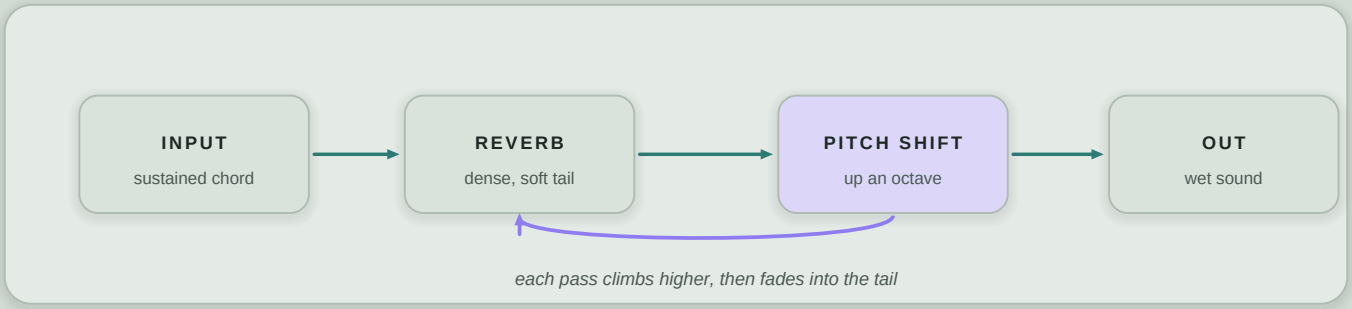
The dark display in the middle of PRISM shows what the plugin is doing. The **glowing ribbon** is your real output waveform of the last couple of seconds. Every grain PRISM starts **blooms as a soft light**: its height shows the stereo position, its size the grain length, and its color the pitch (teal for original, rose for pitched up, violet for pitched down). Behind it, **aurora curtains** swell and brighten with your sound and settle to a faint glow in silence. The slim bars at the sides are the input meter (left) and output meter (right).

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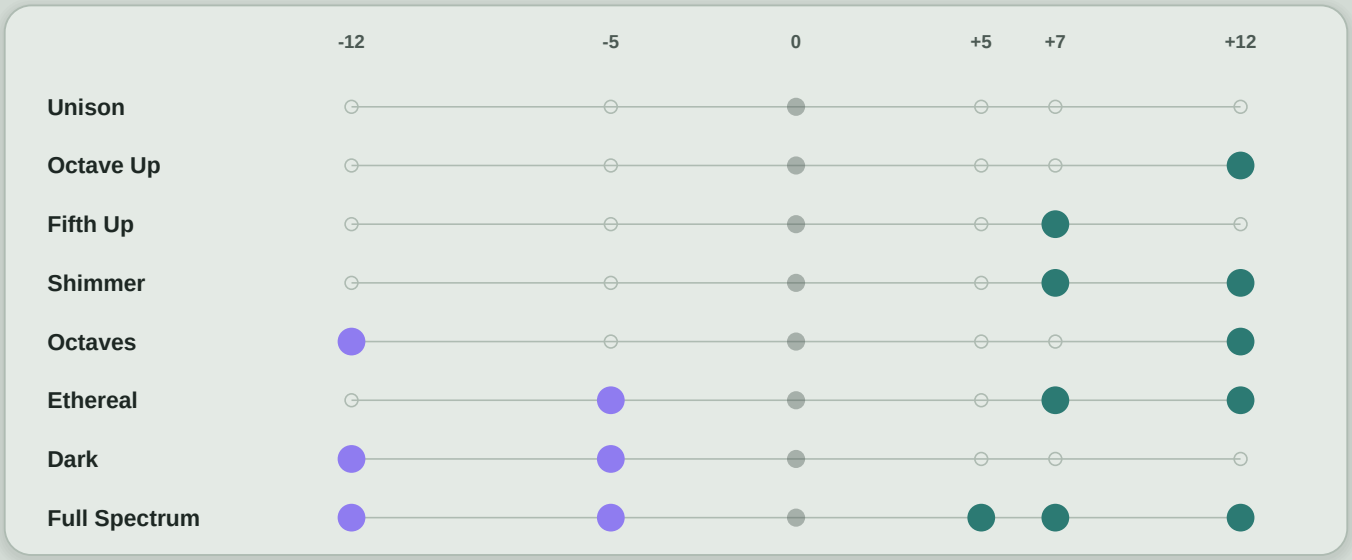
GLOW

Shimmer and Pitch Mode

PRISM makes shimmer in two ways at once. A share of the grains jumps to other pitches, and the reverb's own output is pitch-shifted and fed back into it. Each trip around the loop climbs a little higher before melting into the tail, which is how the glow keeps rising inside the space.



Pitch Mode: which pitches the shimmer can reach



Numbers are semitones. 0 is your original pitch, 12 is one octave up. Teal dots go up, violet dots go down.

SHIMMER sets how much: it is the share of grains that change pitch, and also how strongly the pitched reverb returns to the loop. In the loop the highest upward pitch of the chosen mode is used (Dark uses its downward pitches). **Unison** turns the shimmer off.

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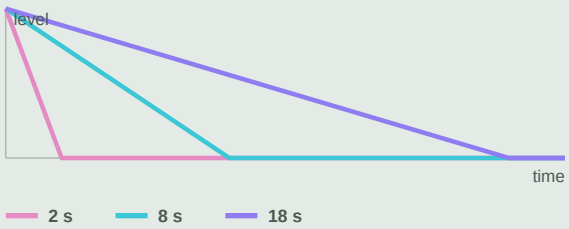
REVERB

The space

PRISM's reverb is a dense 16-line network with a short pre-delay and a set of early reflections that describe the size of the room. Four controls shape how the tail behaves: **Decay**, **Halo**, **Damping** and **Ducker**, with **Pre-Delay** and **Size** to place it.

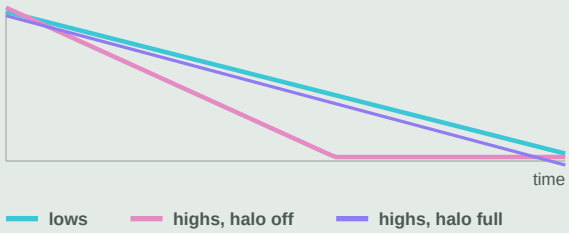
DECAY

0.5 to 20 seconds. How long the tail lasts.



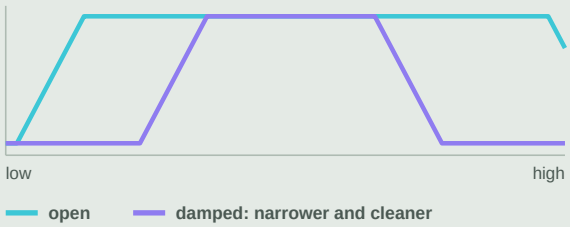
HALO

0 to 100%. How long the highs of the tail last. Low is dark and natural, high is open and airy.



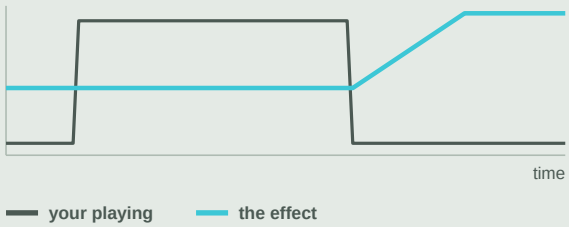
DAMPING

0 to 100%. Tidies the tail: closes its highs and clears its lows.



DUCKER

0 to 100%. Dips the effect while you play (up to about 13 dB), then lets it bloom back.



Size and Pre-Delay

SIZE stretches the room from a small space to a huge hall, and moves the early reflections with it. **PRE-DELAY** adds up to 150 ms before the reverb starts: longer pre-delays keep your source clear and in front of the wash, which is especially useful on vocals and plucked sounds.

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CLOSE-UP

Main controls

The six big knobs shape the character of PRISM. The teal arc shows the value, the white line shows the position, and every number can be typed by double-clicking it.



SPREAD

0 to 100 % · default 80 %

How far back in time grains reach, and how widely they scatter left and right. Low keeps the cloud close and tight. High makes a large, wandering, wide field.

Try 100 % on one sustained note for a huge stereo bed.



SHIMMER

0 to 100 % · default 55 %

How much of the sound climbs to the pitches of the Pitch Mode, and how strongly the pitched reverb returns to the loop.

Try 30 % for a gentle glow, 100 % for a cathedral of octaves.



BLOOM

0 to 100 % · default 35 %

Feeds the finished effect back into the grain tape so the sound swells and smears. Softly limited and always below 100 %, so it never runs away.

Try raising it slowly on short notes.



SIZE

0 to 100 % · default 75 %

How big the room is, from a small space to a huge hall. The early reflections move with it.

Try smaller sizes on drums and plucks, larger on pads and drones.



DECAY

0.5 to 20 s · default 8.5 s

How long the reverb tail lasts. Long decays turn chords into slowly evolving clouds.

Try setting it so each chord clears just before the next one.



MIX

0 to 100 % · default 55 %

The blend between your original sound and PRISM. It uses equal-power crossfading, so the level stays steady as you turn it.

Try 100 % on a send or bus, 20 to 50 % as an insert.

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CLOSE-UP

Reverb detail, pitch and switches

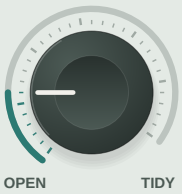


HALO

0 to 100 % · default 55 %

Opens the top end of the reverb tail, from dark and natural to open and airy. The highs of the tail last longer as you turn it up.

Try more halo with more shimmer for an angelic top.



DAMPING

0 to 100 % · default 20 %

Tidies the tail: it closes the top end and clears the lows of the reverb only, so your grains keep their sparkle.

Try more damping when the tail gets in the way of a busy mix.

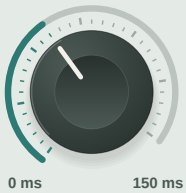


DUCKER

0 to 100 % · default 35 %

Lowers the effect while you play (up to about 13 dB at full), then lets it bloom back when you stop. Your source stays clear and in front.

Try 50 % or more on vocals and lead lines.



PRE-DELAY

0 to 150 ms · default 24 ms

The gap before the reverb starts. Longer gaps keep your source clear and in front of the wash.

Try 60 to 90 ms on vocals and plucked sounds.

PITCH MODE

Shimmer

Chooses which pitches the shimmer can reach: Unison, Octave Up, Fifth Up, Shimmer, Octaves, Ethereal, Dark or Full Spectrum. *Unison turns the shimmer off.*

SWITCHES

FREEZE | REVERSE | TAIL

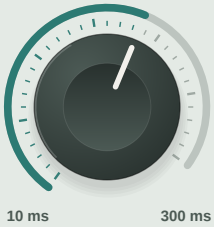
Freeze stops recording so the cloud holds forever. **Reverse** plays some grains backwards. **Tail** lets the sound ring out when you press stop, instead of fading away.

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CLOSE-UP

Detail strip

The bottom strip holds the controls you set once and rarely touch again. They fine-tune the texture, the tone and the movement.



GRAIN SIZE

10 to 300 ms
default 140 ms

Length of each grain. Short grains sound glittery and textured, long grains smooth and washy.



DENSITY

0 to 100 %
default 62 %

How thickly grains overlap, from 3 to 16 at once. Higher is smoother and silkier.



LOW CUT

20 Hz to 1 kHz
default 120 Hz

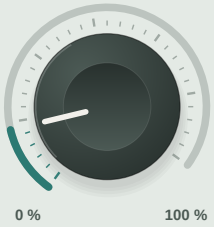
Removes low frequencies from the effect so it stays clean and airy and leaves room for your bass.



HIGH CUT

2 to 20 kHz
default 14 kHz

Removes high frequencies from the effect. Lower it to soften the shimmer.



DARKEN

0 to 100 %
default 15 %

A soft tilt: warmer lows and gentler highs, so nothing sounds harsh.



MOD RATE

0.02 to 5 Hz
default 0.25 Hz

Speed of the slow movement: grain position and pitch drift, and the stereo picture breathing wider and narrower.



MOD DEPTH

0 to 100 %
default 35 %

How strong that movement is. At zero the sound is static. Higher adds drift, width motion and gentle level swells.



OUTPUT

-24 to +12 dB
default +1 dB

Final output level. Presets are level-matched, so you can switch between them at a steady volume.

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STARTING POINTS

Factory presets

1 ANGEL CHOIR — Shimmer

Wide, bright and glowing. The classic ethereal pad.

SHIMMER	55 %
SIZE	75 %
DECAY	8.5 s
BLOOM	35 %
HALO	55 %

2 FROZEN PAD — Octaves

Long, cold, held-in-ice sustain with slow movement.

SHIMMER	30 %
SIZE	85 %
DECAY	14.0 s
BLOOM	55 %
HALO	45 %

3 DARK CLOUD — Dark

Low and heavy, with the highs rolled off.

SHIMMER	20 %
SIZE	80 %
DECAY	11.0 s
BLOOM	45 %
HALO	15 %

4 SOFT SHIMMER — Fifth Up

A gentle fifth and octave glow, subtle on keys.

SHIMMER	45 %
SIZE	55 %
DECAY	5.5 s
BLOOM	25 %
HALO	65 %

5 CELESTIAL BLOOM — Ethereal

Slow-swelling, feedback-rich and full of shimmer.

SHIMMER	60 %
SIZE	90 %
DECAY	13.0 s
BLOOM	62 %
HALO	60 %

6 GLASS CATHEDRAL — Octave Up

Huge reverb with crisp grain sparkle.

SHIMMER	50 %
SIZE	95 %
DECAY	16.0 s
BLOOM	30 %
HALO	70 %

7 NORTHERN DRIFT — Ethereal

Slow, wide movement with a wandering pitch.

SHIMMER	40 %
SIZE	70 %
DECAY	9.0 s
BLOOM	40 %
HALO	50 %

8 VELVET NEBULA — Dark

Warm, dark and wide.

SHIMMER	25 %
SIZE	85 %
DECAY	12.0 s
BLOOM	50 %
HALO	20 %

9 MOONLIT HALO — Fifth Up

Quiet and airy. Lovely on vocals and guitars.

SHIMMER	50 %
SIZE	65 %
DECAY	7.0 s
BLOOM	20 %
HALO	60 %

10 INFINITE HEAVEN — Full Spectrum

Maximum size and endless decay.

SHIMMER	65 %
SIZE	100 %
DECAY	19.0 s
BLOOM	75 %
HALO	60 %

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IDEAS

Recipes and workflow

Presets are starting points. These recipes show how to bend them to common jobs. Always set **MIX** by ear, in context with the rest of your track.

1 WARM KEYS BLOOM

Preset: Angel Choir

MIX 35 to 45 %, LOW CUT 200 Hz, DUCKER 40 %.

The low cut keeps the wash out of the bass, and the ducker lets each chord speak before the cloud blooms.

2 GUITAR AMBIENCE

Preset: Moonlit Halo

MIX 30 %, SHIMMER 30 %, DECAY 6 s.

A quiet halo that sits behind the playing without covering the attack.

3 VOCAL HALO

Preset: Soft Shimmer

MIX 20 to 25 %, PRE-DELAY 60 to 90 ms, HIGH CUT 10 kHz, DUCKER 50 %.

The pre-delay and ducker keep the words clear, and the shimmer opens up between phrases.

4 ENDLESS DRONE

Preset: Frozen Pad

Play a chord, switch FREEZE on, then stop playing. Raise SIZE and DECAY.

The cloud holds the chord as a continuous bed you can play over.

5 REVERSE SWELLS

Preset: Celestial Bloom

REVERSE on, GRAIN SIZE 250 ms, SPREAD 100 %, BLOOM 50 %.

Backward grains build into slow swells that lead into each new chord.

6 SOUND DESIGN

Preset: Infinite Heaven

PITCH MODE Full Spectrum on a single held note. Automate SHIMMER and SIZE.

One note becomes an evolving landscape. Add MOD DEPTH for slow stereo movement.

Mixing tips

- Use PRISM on a **send or bus** at 100 % MIX when several instruments should share the same space.
- Keep the **LOW CUT** above 100 Hz when the effect gets in the way of the bass.
- **Automate** SHIMMER, SIZE or FREEZE for builds and transitions. Every control is automatable.
- If the top gets too bright, lower **HIGH CUT** or **HALO**, or raise **DARKEN**.

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REFERENCE

Install, support and credits

Installing

PRISM installs as an Audio Unit for Logic Pro, a VST3 for FL Studio and other hosts, and a Standalone app. It needs **macOS 12 or later** and runs natively on Apple Silicon and Intel Macs.

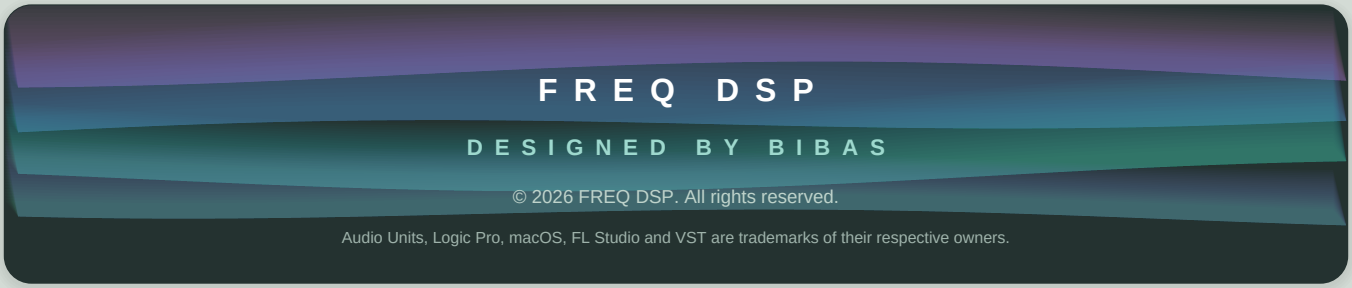
Format	Location
Audio Unit (Logic Pro)	~/Library/Audio/Plug-Ins/Components/PRISM.component
VST3 (FL Studio and others)	~/Library/Audio/Plug-Ins/VST3/PRISM.vst3
Standalone	The PRISM app, opened on its own

Troubleshooting

Problem	What to try
PRISM is not in Logic Pro	Quit and reopen Logic. If it is still missing, open Logic Pro, Settings, Plug-in Manager, select PRISM and choose Reset and Rescan Selection.
PRISM is not in FL Studio	Open Options, Manage Plugins, then Find plugins, and tick PRISM in the list.
The sound cuts off when I stop playback	Switch on TAIL under the display. By default PRISM fades out quickly when the host stops.
The top end is too bright or harsh	Lower HIGH CUT or HALO, or raise DARKEN or DAMPING. Reduce SHIMMER if needed.
The effect covers my source	Raise DUCKER, lengthen PRE-DELAY or lower MIX.
Levels jump between presets	Presets are level-matched. Check OUTPUT and MIX if you changed them.

Technical notes

Stereo in, stereo out. The dry path adds no latency. Memory is allocated when the plugin loads and never while audio is playing, and the Bloom and Shimmer loops are softly limited so they cannot run away. A gentle safety limiter sits just below full scale. All controls are smoothed to avoid zipper noise and can be automated. Settings, preset and theme are saved with your project.



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